

# Simulated-First Bench Rover

Observe, decide, stop, then permit bounded motion

*A stopped intent is software evidence. Removed motor power is the independent stop.*

|                      |                                                   |               |                                           |
|----------------------|---------------------------------------------------|---------------|-------------------------------------------|
| <b>Lesson</b>        | 021 — integration project                         | <b>Time</b>   | 240–360 minutes                           |
| <b>Prerequisites</b> | Lessons 001–020                                   | <b>Board</b>  | Arduino Mega 2560                         |
| <b>Evidence</b>      | host replay, six direction/enable lamps, and D13  | <b>Status</b> | host verified; E1 and E2 bench cards open |
| <b>Safety</b>        | E1 indicator simulation; E2 continuation deferred |               |                                           |

**Safety checkpoint.** Do not place a rover on the floor during initial work. Complete the host replay and resistor-limited LED simulation before connecting a motor driver. Select one exact driver and motor only after checking their primary datasheets, supply limits, stall current, thermal limits, flyback strategy, and truth table. A physical switch must remove motor-supply power. The software stop button does not replace it. No person, animal, sharp tool, hot item, or hazardous load may ride on the rover. Keep away from roads, stairs, hair, loose clothing, and unattended operation.

## 1 Mission

Compose range, wheel evidence, bounded motor intent, a static route, and an operator stop into a deterministic supervisor:

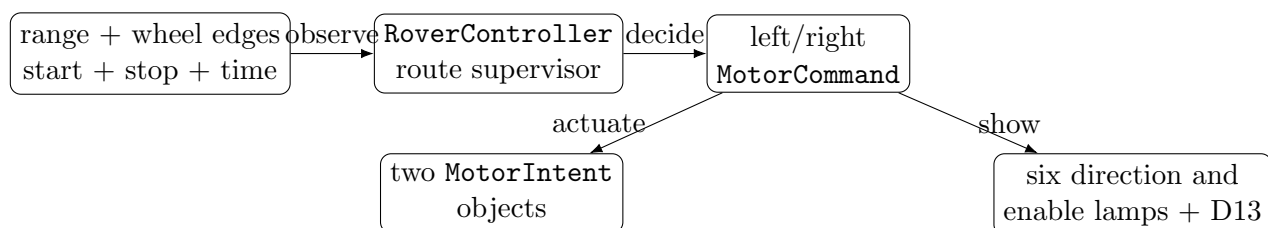
$$\{\text{range, wheel edges, start, stop, time}\} \rightarrow \text{RoverController} \rightarrow \{\text{left intent, right intent, mode}\}.$$

The first rover is a host model. The second is six resistor-limited lamps that show the two motor-driver interfaces without motor power. Only the final, still-unaccepted stage connects a documented driver and raises the wheels clear of the bench. These stages test the same route and supervisor.

By the end, you can:

- explain why the supervisor emits intent but owns no pins or clock;
- replay the same inputs and obtain the same outputs byte for byte;
- show that stop dominates every driving and turning state;
- distinguish command evidence, electrical evidence, and physical motion;
- diagnose stale range, no initial motion, wheel disagreement, route timeout, and operator stop without relying on Serial;
- leave physical acceptance open without inventing a successful bench result.

## 2 Composition and ownership



**RoverController** is hardware-neutral. It receives complete observations and produces complete intent. The canonical Mega sketch owns seven LED components, two **MotorIntent** objects, an **UltrasonicRanger**, and the controller. Its range echo and wheel edges are deterministic simulations. Setup acquires every object before admitting commands; shutdown first emits stopped intent, then makes the evidence outputs inactive and releases them in reverse order.

| Layer                  | Owens                                              | Does not claim                                     |
|------------------------|----------------------------------------------------|----------------------------------------------------|
| <b>RoverController</b> | bounded route copy and deterministic state         | pins, timers, clock, motors                        |
| <b>MotorIntent</b>     | bounded applied command and reversal<br>dead time  | pins, mechanical motion, or safe supply<br>removal |
| Mega sketch            | simulated observations and seven LED<br>components | hidden control policy                              |
| Physical disconnect    | interruption of motor-supply energy                | software state                                     |

### 3 Supervisor contract

Lessons 019 and 020 provide the landed **UltrasonicRanger** and **MotorIntent** dependencies. The exact controller boundary is:

```

struct RoverInput
{
    RangeReading range;
    TimePoint rangeObservedAt;
    RoverWheelObservation leftWheel;
    RoverWheelObservation rightWheel;
    bool startEvent;
    bool stopRequested;
    TimePoint observedAt;
};

struct RoverControllerSnapshot
{
    MotorCommand leftMotor;
    MotorCommand rightMotor;
    RoverMode mode;
    RoverStopReason stopReason;
    Status status;
    RangeReading range;
    uint8_t routeIndex;
    bool stopDominated;
};

Status      initialize () noexcept;
Status      update      (const RoverInput& input) noexcept;
void        shutdown    () noexcept;
RoverControllerSnapshot snapshot () const noexcept;

```

The route contains only fixed records: drive, turn left, turn right, pause, or finish; target wheel edges; maximum duration; and bounded duty. No allocation or hidden clock is needed. A caller timestamp, observations, prior state, and configuration determine each output.

#### 3.1 Stop dominance

Evaluate the stop request before route progress. A stop request in **Ready**, **Driving**, **Turning**, or **ObstacleHold** selects zero-duty commands for both motors and enters **Stopped**. A start/stop chord is a stop. Shutdown also

emits stopped intent. Sensor faults, stale range, initial-motion timeout, wheel disagreement, route timeout, unexpected motion, and invalid time all request both motors stopped.

**Invariant to predict:** whenever `stopRequested` is true, both applied commands are **Stopped** in that same update. Stop also dominates a simultaneous start event. The two `MotorIntent` objects apply those commands before the six LEDs present their applied snapshots. A software trace proves the supervisor intent; the enable witnesses prove only the presented logic state. Only a future physical disconnect can prove removal of motor-supply energy.

Restart is deliberate. Releasing stop or recovering a sensor does not resume the route. Require stop release, fresh valid range, stationary evidence, and a new start edge. Reset returns to inactive behavior and never restarts motion.

## 4 Deterministic trace and replay

Use one row per observation:

```
time,range-state,range-mm,left-total,right-total,start,stop-request,
rover-mode,left-command,right-command,route-index
```

The host plant is an integer-only teaching model. It converts commands and elapsed time into synthetic edge totals and scripted obstacle range. It can inject wheel gain mismatch, start delay, stopping delay, and faults. It is not a model of traction, stopping distance, supply sag, driver heating, or a specific chassis.

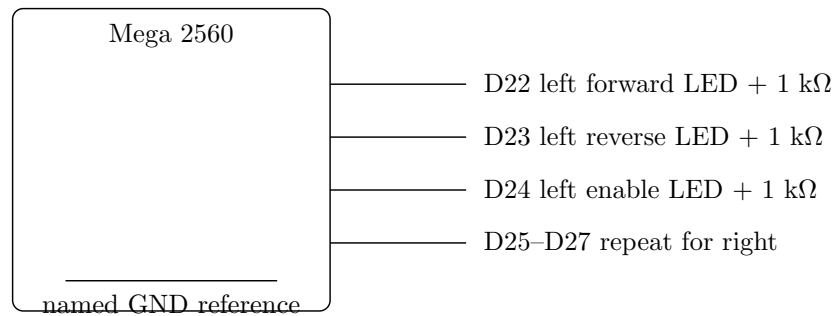
1. Run a fixed route twice with the same configuration and input rows.
2. Compare every output field and serialized trace byte.
3. Insert a stop request into each active state.
4. Repeat at exact timeout boundaries and across timestamp rollover.
5. Change one input only; predict the first output row that may differ.

| Injected observation             | Required decision                   | Circuit-native later evidence                               |
|----------------------------------|-------------------------------------|-------------------------------------------------------------|
| fresh clear range, matched edges | continue bounded route step         | expected direction lamps and enable waveform                |
| range at stop threshold          | obstacle hold, both duties zero     | both enable witnesses inactive                              |
| stale or invalid range           | sensor fault, both duties zero      | red fault cadence; both enable witnesses inactive           |
| one wheel supplies no new edges  | motion fault after configured bound | red fault state in the snapshot; all six witnesses inactive |
| stop request                     | stopped immediately                 | all six witnesses inactive                                  |

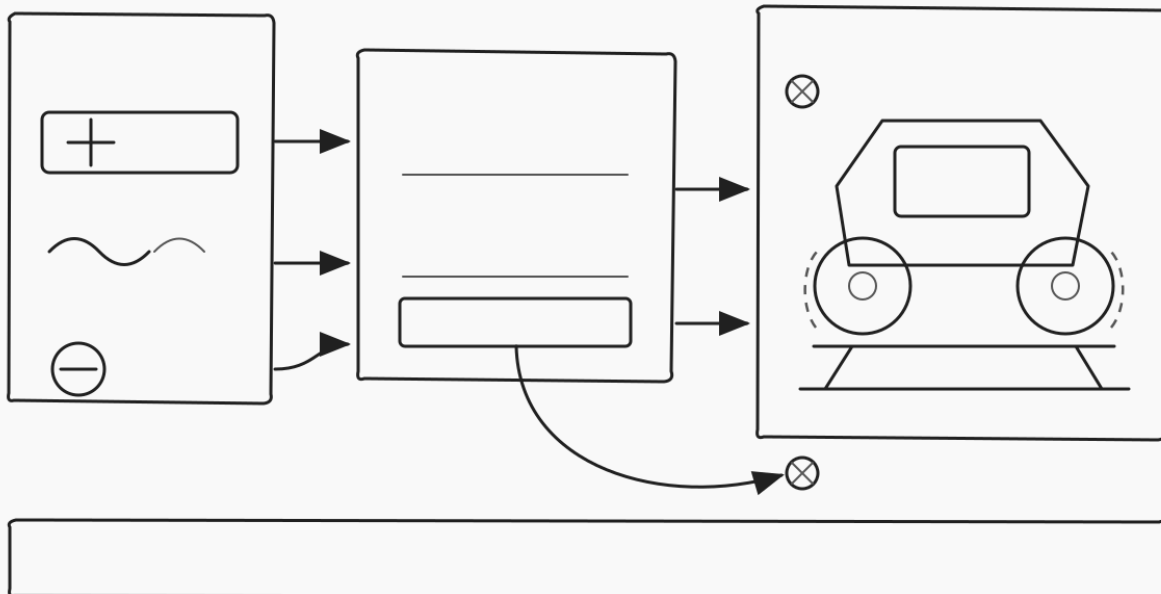
Serial may preserve or compare this trace. It never commands movement and is never the only proof.

## 5 Stage A: inert motor-interface simulator

Build this before selecting a driver. Replace each driver's forward, reverse, and enable input with one LED and its own 220  $\Omega$  or larger resistor. The six LEDs represent logic intent only.



**Figure description:** a pencil-style orientation view places the Mega to the left and six individually resistor-limited indicator channels to the right. D22–D24 show left forward, reverse, and enable; D25–D27 repeat them for the right. Every cathode returns to the named Mega ground. The built-in D13 LED is separate acquisition evidence. This is an orientation drawing, not an authoritative driver schematic.



**Pencil plate description:** the rover is raised on blocks with both wheels clear as an illustration of the deferred E2 stage. The supported E1 circuit contains only the Mega and logic indicators. Forward, reverse, and enable evidence runs to each side. A hand at the motor-power disconnect emphasizes that software stop and energy removal would be separate observations in a future powered stage.

Use pins selected by the canonical Lesson 021 sketch and its pin table; do not infer real driver wiring from the orientation drawing. At this stage no motor supply, H-bridge, motor, wheel, or battery is present.

1. With USB removed, inspect every resistor and LED polarity.
2. Predict all six lamps inactive at startup.
3. Start the route and observe direction plus nonzero-duty enable intent.
4. Request a reversal and observe the dead-time dark interval.
5. Press software stop in every active mode; both enable lamps must become inactive immediately.
6. Call shutdown; then measure D22–D27 against the named GND according to the endpoint's documented high-impedance procedure.

## 6 Circuit-native observation plan

The circuit explains itself through independent channels:

| Question                           | Observation                                        | What it proves                                      |
|------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| Did setup acquire its objects?     | built-in D13 turns on after all objects initialize | the acquisition sequence completed                  |
| What command reached each adapter? | forward/reverse/enable lamps                       | applied direction and nonzero-duty intent           |
| Are enable signals inactive?       | D24 and D27 relative to named logic GND            | electrical logic state, not removal of motor energy |
| Did shutdown release the outputs?  | D22–D27 measured against named logic GND           | high impedance after stopped intent was presented   |

The six witnesses deliberately encode commands rather than every controller mode. A host snapshot distinguishes obstacle hold, fault, route completion, and operator stop when each correctly presents the same all-dark output.

## 7 Stage B: driver boundary, no wheels on the surface

**Safety checkpoint.** This continuation is blocked until one exact H-bridge module and motor are identified from primary documentation. Do not generalize a pin table between modules that merely look alike. Do not power a motor from Mega 5 V or from a GPIO pin.

Complete this worksheet before wiring:

| Item                       | Recorded value/source | Item                           | Recorded value/source |
|----------------------------|-----------------------|--------------------------------|-----------------------|
| driver exact part/revision | _____                 | motor exact part               | _____                 |
| logic supply/range         | _____                 | motor supply/range             | _____                 |
| continuous/peak current    | _____                 | measured/limited stall current | _____                 |
| flyback strategy           | _____                 | thermal limit/handling         | _____                 |
| input truth table source   | _____                 | physical disconnect rating     | _____                 |
| common-reference/isolation | _____                 | back-power prevention          | _____                 |

The authoritative schematic must show logic supply, motor supply, return or isolation boundary, protection, driver outputs, both motors, named left/right enable test points, and the physical motor-power disconnect. Its exact wiring belongs to the selected module's accepted bench card; this draft intentionally supplies no generic H-bridge pinout.

### 7.1 Wheels-raised sequence

1. Remove USB and motor power. Raise and secure the chassis so no wheel can touch the bench. Remove fan blades or other attachments.
2. Open the physical motor-power disconnect. Inspect against the exact schematic and verify polarity and shared-reference/isolation policy.
3. Apply logic power only. Confirm both enable test points inactive during startup, reset, stop, fault, and shutdown.
4. Close motor power at the lowest documented current-limited setting. Test one motor at the lowest useful duty for the shortest route step.

5. Observe direction indicator, the named enable test point, a wheel mark, and the selected encoder evidence together. Disagreement is a stop condition.
6. Test stop, stale range, blocked echo, one stalled wheel observation, and reset. Open the physical disconnect after every test.

Do not infer safe floor behavior from a raised-wheel test. Traction, inertia, stopping distance, chassis balance, radio loss, and environmental obstacles have not been accepted.

## 8 Predict, observe, interpret

| Prediction                           | Observation                                                                 | Interpretation                                                                |
|--------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| startup cannot move                  | both enable test points inactive; all direction lamps inactive              | supports safe logic state; does not prove motor supply is removed             |
| turn requests opposite wheel intents | direction lamps differ; wheel marks move as predicted                       | supports adapter and raised-wheel direction; does not prove floor trajectory  |
| near obstacle dominates route        | obstacle cadence appears and both enable signals become inactive            | supports range-to-stop chain when the range sample is fresh and valid         |
| stop dominates every state           | stop indication appears and both enables become inactive in the same update | supports software stop and logic output; physical disconnect remains separate |
| shutdown releases outputs            | zero duty precedes documented high impedance                                | supports lifecycle order when measured at both enable pins                    |

## 9 Diagnosis without Serial

| Observation                             | Likely boundary                             | Next unpowered check                                                           |
|-----------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------|
| acquisition walk stops early            | resource/configuration conflict             | compare pins, timers, and initialization order                                 |
| mode says drive; enable lamp dark       | motor adapter or resource failure           | compare output snapshot, enable pin, resistor, and LED polarity                |
| enable lamp pulses; wheel mark still    | load power, driver, motor, or mechanics     | open disconnect; inspect supply, driver rating, shaft freedom, and connections |
| wheel moves; encoder LED still          | marker, sensor gap, input, or capture fault | inspect marker alignment and sensor wiring                                     |
| one direction LED overlaps its opposite | adapter fault                               | open disconnect immediately; no motor test until corrected                     |
| fault clears without new start          | supervisor defect                           | preserve trace; return to host replay before any powered continuation          |

## 10 Acceptance record

### Host and inert-indicator evidence

| Check                                                  | Result                        |                               | Evidence |
|--------------------------------------------------------|-------------------------------|-------------------------------|----------|
| same trace twice is byte-identical                     | <input type="checkbox"/> pass | <input type="checkbox"/> fail | _____    |
| stop from every active state gives two zero duties     | <input type="checkbox"/> pass | <input type="checkbox"/> fail | _____    |
| stale/invalid range and encoder faults stop intent     | <input type="checkbox"/> pass | <input type="checkbox"/> fail | _____    |
| LED reversal shows documented dark interval            | <input type="checkbox"/> pass | <input type="checkbox"/> fail | _____    |
| startup, reset, fault, shutdown leave D22–D27 inactive | <input type="checkbox"/> pass | <input type="checkbox"/> fail | _____    |

### E2 bench card — open until physically performed

Board/serial: \_\_\_\_\_ Date: \_\_\_\_\_ Reviewer: \_\_\_\_\_

Driver/motor revisions: \_\_\_\_\_

Supply/current limit: \_\_\_\_\_

Instruments: \_\_\_\_\_ Physical disconnect: \_\_\_\_\_

- ☐ exact schematic and primary datasheets attached;
- ☐ wheels raised and chassis restrained;
- ☐ motor stall-current boundary recorded without exceeding any driver, supply, wire, or connector rating;
- ☐ startup, reset, stop, hold, each fault, shutdown, and power removal measured at both enable test points;
- ☐ load power physically removed during loss-of-control test;
- ☐ no unexpected movement, heat, odor, reset, or back-powering.

Measured observations/deviations:

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**Release status:** this lesson remains a **host verified; E1 indicator and E2 physical acceptance open**. The dependencies, canonical example, deterministic tests, and Mega compilation have landed. The completed bench card must still record real instruments, supplies, pins, observations, reviewer, and date. Compilation, simulation, LEDs, and this worksheet are not physical acceptance.

## 11 Further experiments

1. Minimize one replay that enters each fault using the fewest rows.
2. Vary the host plant's left-wheel gain and find the exact configured mismatch boundary. Explain why it is not a physical tolerance claim.
3. Change the existing route pause. Verify that synthetic wheel movement during the pause becomes explicit evidence rather than silently advancing the route.
4. Design a separate stop indicator, then explain what evidence it adds beyond the six all-dark command witnesses.
5. Propose a floor-test enclosure and stopping-distance protocol, but do not perform it until a separate reviewed acceptance plan exists.

## 12 CLI route

```
make host-test
make host-test-sanitize
make arduino-Lesson021BenchRover
make size-check
make lessons
make lessons-check
make upload-Lesson021BenchRover PORT=/dev/ttyACM0
make serial-log PORT=/dev/ttyACM0 SERIAL_LOG=build/lesson021.csv
make monitor PORT=/dev/ttyACM0 BAUD=115200
```

The upload, monitor, and log targets support the lesson; no IDE is required. Serial logging adds context to the non-Serial evidence and never controls motion.

## 13 Primary references

- Canonical Mega example
- `RoverController` interface
- `UltrasonicRanger` interface
- `MotorIntent` interface
- Deterministic rover tests
- Ranger tests and motor-intent tests
- Arduino Mega 2560 Rev3 hardware documentation: <https://docs.arduino.cc/hardware/mega-2560>
- Arduino built-in Ping range example: <https://docs.arduino.cc/built-in-examples/sensors/Ping/>
- Arduino Motor Shield Rev3 documentation, as a representative boundary rather than the wiring authority for an unidentified module: <https://docs.arduino.cc/hardware/motor-shield-rev3/>
- ADK safety model: `docs/SAFETY_MODEL.md`
- ADK executable rover design: `docs/design/LESSONS_019_021_ROVER.md`